SEPTA SWARTHMORE STATION DISPLAY

Simple Case Build Guide

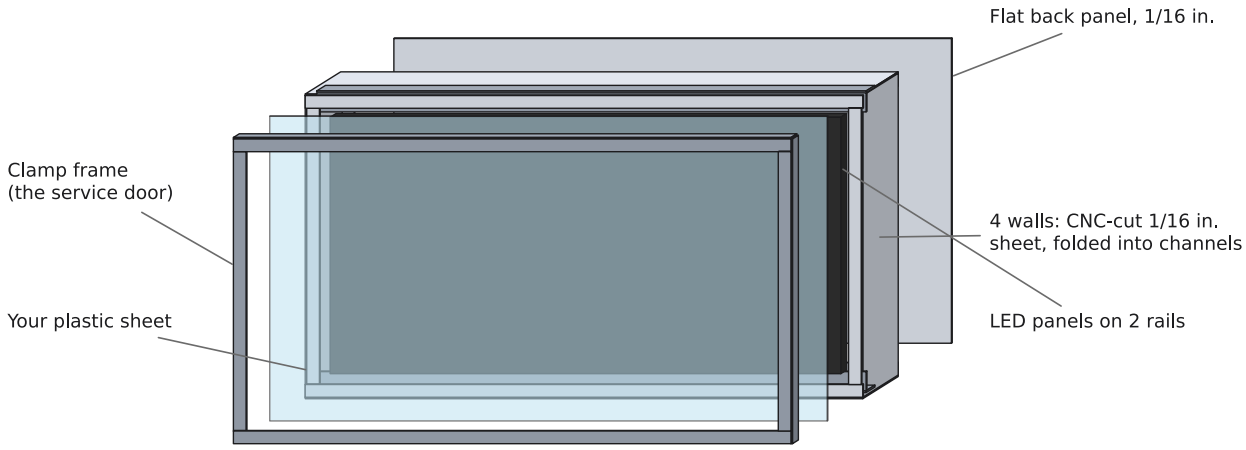
Revision 2: a light box made from the 1/16 in. aluminum you already have, cut and scored on the CNC, folded by hand, riveted and sealed, with printed vents, a commercial TV mount, and your existing plastic sheet as the window.

≈ $403
parts, with an outdoor TV mount. About $244 with the Z-bar instead. Aluminum not counted.

≈ 14 h
build time, plus CNC time and printing

CNC + hand tools
CNC router, 3D printer, rivet tool, caulk gun

2 people
to hang it (about 50 lb)

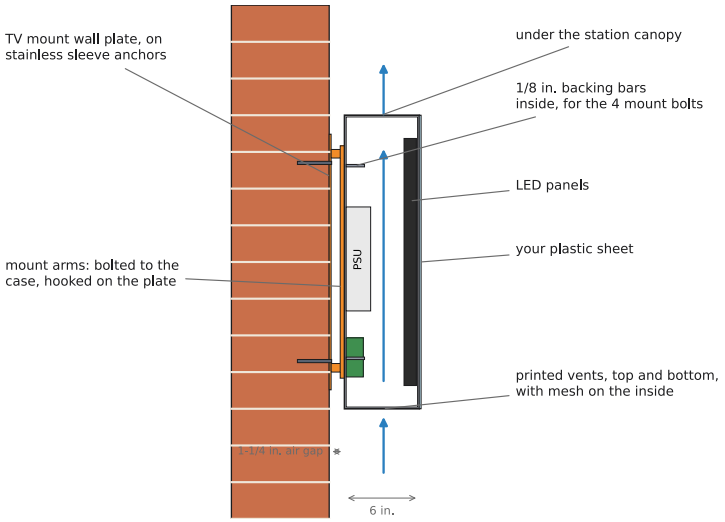


Before you drill the station

The building belongs to SEPTA, so mounting needs their written OK first, and they may want their own crew to do the drilling or the mains connection. The outlet is an electrician's job. Everything else in this guide is a normal workshop job.

How it works

- **Box:** four walls cut from 1/16 in. sheet and folded into channels, with a 1 in. lip along each long edge. The folds make thin sheet stiff. A flat back panel is riveted to the back lips and sealed, and stays on for good.
- **CNC:** every rivet pilot, vent window and fold line is cut in one setup, so the parts line up without measuring.
- **Window:** your existing plastic sheet sits on foam on the front lips and is held by a **clamp frame**, screwed into rubber **well nuts**. Removing those screws is how you open the case.
- **Inside:** the LED panels bolt to two rails across the top and bottom. The PSU and Pi screw to the back.
- **Air:** printed louvres in the bottom and top walls, with mesh on the inside, let warm air rise straight through.
- **Wall:** a commercial outdoor TV mount. Its plate goes on the brick, its arms bolt to the case, and the arms hook on and lock.



Choosing a mount

The case carries a 600 × 400 mm VESA bolt pattern on its back, so any of these TV mounts bolts straight on. All carry well over the 50 lb board.

Mount	Rated	Depth	≈ $	Notes
SunBrite SB-WM-F-L-BL (recommended)	165 lb	1-1/4 in.	173	Fixed. Outdoor-rated 3 mm steel, stainless hardware, lockable tabs. Takes up to 600 × 500. Stock varies; B&H lists it as a special order.
Furrion F2AA002ABBK	110 lb	about 1.4 in.	120	Saves about $55. Powder-coated galvanized steel, built-in anti-theft, but it tilts 0 to 12°, so set it to 0° and check it stays there.
Hangman HM-30D Z-bar	300 lb (retailer)	1/2 in.	14	The budget route; see the box at the end of the hanging section. Not made for a public site, so it needs anti-lift brackets.

> **The mount does not fix the wall**
>
> TV mounts assume wood studs or their own hardware. On this brick, use the stainless sleeve anchors in this guide, and check the slots on the plate before buying anchors. Confirm with SEPTA what they allow in the wall.

Three decisions, and why

Joining the walls, back and front

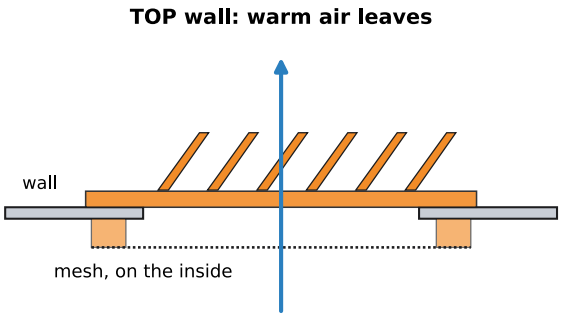
Method	Good	Bad	Use it for
Closed-end rivets + silicone	No heat, no warping, quick, cheap. The closed end has no path for water through the rivet.	Permanent. The seal depends on the silicone between the parts.	**Walls and back**
TIG welding	One solid, fully sealed box.	1/16 in. aluminum takes real skill and tends to warp. Heat softens the metal near the weld. A sealed box is not the goal, because this case has vents.	Not recommended
Screws into well nuts + foam	Comes apart, so you can always get in. The rubber seals each screw.	More hardware, slower.	**The front**
Structural tape or adhesive	Neat, no holes.	Needs careful surface prep, and is hard to check once closed.	Not on its own

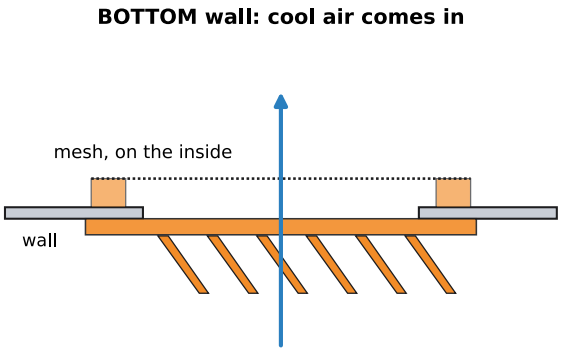
Vents: yes, mesh on the inside

Put the printed louvres outside and the mesh behind them. The slats keep falling water off the mesh, and being inside keeps it out of the weather.

Freezing is not a danger to the board. Vents can only ice up when it is below freezing outside, which is exactly when the board needs the least cooling. If one ices over, the box just runs a little warmer until it melts. The dangerous case is a blocked vent in July, so clean the mesh each spring.

- Use ordinary aluminum insect screen (about 1.5 mm openings), not fine mesh. Fine mesh holds a film of water that freezes.
- Print in **ASA**. PETG is fine in the shade. Never PLA: it softens around 55 °C and cracks in the cold.
- Screw the vents in so they come out for cleaning.
- Drill four 1/8 in. weep holes in the bottom wall, because the lips make it a shallow tray.



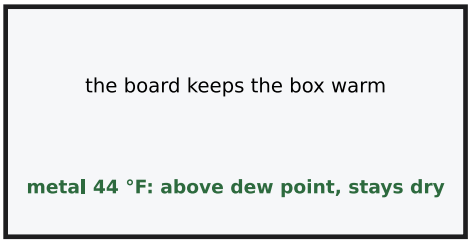


Printed ASA louvres outside, mesh behind them, held by a printed frame. 8 in. long so it prints on an ordinary bed. Slats tilt so nothing falling can land on the mesh. Water on the bottom vents simply runs out.

Humidity: worth caring about, and easy to manage

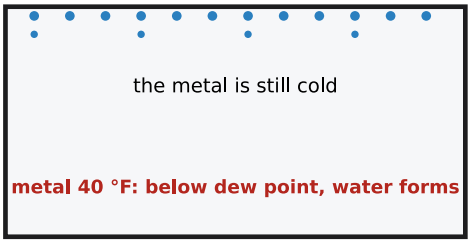
Damp air does little harm on its own. The problem is **liquid water**, which forms when a surface is colder than the air's dew point.

Board running



example: outside 36 °F, dew point 32 °F

Off overnight, then a warm damp morning



example: outside 62 °F, dew point 58 °F

Water only forms on a surface colder than the air's dew point. So keep it powered, and let it breathe.

Do

- **Leave it powered, all year.** The board's own heat keeps surfaces above the dew point. Overnight dimming is fine.
- **Keep the vents open.** A sealed box traps moisture every time it warms and cools. In a field test by Gore, a vent maker, a sealed box with desiccant had serious condensation by day 4, and the vented box shed moisture much better.
- **Fit drip shields** over the PSU and Pi, bent from scrap, so a drop from the top wall lands on metal.
- **Run every cable in from below,** with a drip loop.
- **Conformal-coat the Pi and bonnet** (step 12) as a backstop.

Do not

- Seal the vents for winter. It makes condensation worse.
- Add desiccant. In a vented box it saturates in days.
- Cold-start after days unplugged on a damp morning. Open the frame and let it dry first.
- Coat connectors, the SD slot, the USB ports, the fan or the header pins. Mask them first.

> **Heat and cold**
> Expect the inside to run about 5 to 10 °C above outside. In a 38 °C heat wave, that is the top of the Pi's 50 °C rating, so keep it out of direct sun and keep the Active Cooler.

Before you buy: measure

Every size comes from your plastic sheet. The example numbers are for a **44 × 24 in.** sheet; fill in your own.

Measure	Example	Yours	Why
W: plastic sheet width	44 in.		Sets the case width
H: plastic sheet height	24 in.		Sets the case height
Panel array width	40.3 in.		At least 2 in. less than W
Panel array height	20.2 in.		At least 2 in. less than H
PSU size	about 8.5 × 4.5 × 2 in.		Must fit behind the panels
Sheet thickness, with calipers	0.063 in.		Sets the score depth
CNC bed size	48 × 96 in.		Must take a 46 × 26 in. part
Is there a sheet-metal brake at the shop?			Better than scoring, if so

If the CNC bed is smaller than 46 × 26 in.
Cut the back and long walls in two setups: drill two 1/4 in. dowel holes through the sheet into the spoilboard, machine half the part, slide it over, and re-register on the dowels.

Flat sizes, in inches

Part (all 1/16 in. sheet)	Formula	Example	Yours
Back, flat	W × H	44 × 24	
Top and bottom walls ×2	W × 7-3/4	44 × 7-3/4	
Side walls ×2	(H – 1/8) × 7-3/4	23-7/8 × 7-3/4	
Clamp frame, top and bottom ×2	W × 2-3/8	44 × 2-3/8	
Clamp frame, sides ×2	(H – 2) × 2-3/8	22 × 2-3/8	
Corner pieces ×4	4 × 1-7/8	4 × 1-7/8	
Drip shields ×2	10 × 3 and 5 × 3	from spare	

Score lines are 15/16 in. in from each long edge of the walls, and on the frame 15/16 in. from the edge that will face front. Change the 7-3/4 only if you change the 6 in. depth: it is depth – 1/8 + 1-7/8.

Shopping list

Prices from retailer listings in October and November 2026; check before ordering. Aluminum and a new window sheet are listed but **not counted**.

Parts (counted)

Item	Qty	Where	≈ $
Outdoor fixed TV mount, SunBrite SB-WM-F-L-BL	1	BBQGuys, or see "Choosing a mount"	173
Stainless sleeve anchors, 1/4 in., about 2-1/4 in. long, with fender washers	6	Home Depot	14
Closed-end aluminum rivets, 5/32 in., grip to cover 0.125 and 0.188 in. (100)	1	McMaster-Carr; read the grip range	25
Clear outdoor silicone, 10.1 oz (neutral cure)	2	GE Silicone 2, Home Depot	22
Rubber well nuts, #10-32	12	Bolt Depot	24
Pin-in Torx security screws, #10-32 × 1/2 in., stainless, 25-pack	1	Albany County Fasteners	20
EPDM foam tape, 1/8 × 1/2 in. × 65 ft	1	Okayasu, Amazon	15
Cable gland, PG13.5 nylon (pack)	1	Baomain, Amazon	10
Nylon M2.5 standoff kit (for the Pi)	1	Amazon, search "M2.5 nylon standoff kit"	9
M8 × 20 mm stainless bolts, nylon lock nuts, washers (VESA)	4 sets	Hardware store	6
Stainless fasteners: M4 × 10 mm ×4 (PSU), #10-32 screws, lock nuts, star washer (earth bolt)	1 set	Home Depot, Everbilt stainless drawers	12
Aluminum insect screen (about 3 sq ft)	1	Hardware store	6
Green 14 AWG wire (2 ft) and two ring terminals	1 set	Hardware store	6
ASA filament, 1 kg (skip if the shop has it)	1	Any filament seller	31
Conformal coating, MG Chemicals 419D aerosol	1	MG Chemicals, or DigiKey	30
Parts total			**≈ 403**

Tools

Tool	Used for	≈ $ if bought
1/8 in. single-flute O-flute carbide bit, e.g. Amana 43530	all through-cuts	41
90° zero-point carbide V-bit rated for aluminum, e.g. Amana A45614. Not a flat-tip bit: flat tips cut too wide for 1/16 in.	scoring the folds	40
MDF spoilboard, screws, WD-40, safety glasses	holding the sheet, lubricating	shop
Two 4 ft lengths of steel angle or hardwood, C-clamps	homemade folding bar	shop

Tool	Used for	≈ $ if bought
Drill; #20 and 3/16 in. and 3/8 in. bits; 1/4 in. masonry bit	opening rivet pilots, well nuts, anchors	6
Hand rivet tool (Arrow RT100M or similar)	every joint	22
Hacksaw or mitre saw with a non-ferrous blade; file; deburring tool	the thick bars	shop
T-25 pin-in Torx bit, caulk gun, tape measure, square, level	assembly	11

Aluminum and window (listed, not counted)

Item	Qty	Where	≈ $
1/16 in. aluminum sheet, 48 × 96 in. (or the equivalent in smaller sheets)	1	You already have this	0
Aluminum angle, 1-1/2 × 1-1/2 × 1/8 in., 8 ft	2	Everbilt 802617, Home Depot	104
New window sheet, UV-protected polycarbonate (only if the old one is too small or damaged)	1	TAP Plastics, cut to size	100 to 200

The build

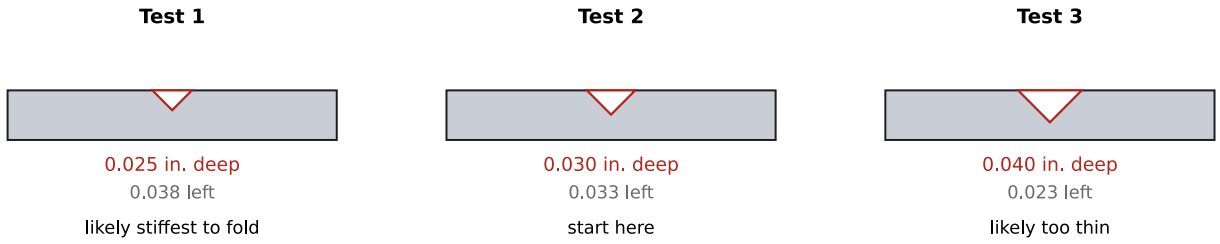
Sixteen steps. In the pictures, **orange** is what that step adds. Label every part with masking tape as it comes off the machine.

parts tools

1 Test the alloy and the score first

45 min

scrap of your sheet | CNC, V-bit | calipers | clamp, bar



Drawn much larger than life. Fold each by hand to 90°, then look at the outside of the corner under a light.

1. Cut three 6 × 4 in. coupons and score each across the middle, at the three depths shown. Take two passes of about 0.015 in. each.
2. Fold each by hand to 90° over a clamped bar, once, score on the inside.
3. Look at the outside of each corner under a bright light. Use the deepest score that folds cleanly and has no cracks.
4. Rivet a scrap to the fold of the best coupon and try to break it by hand, so you know what a good one feels like.

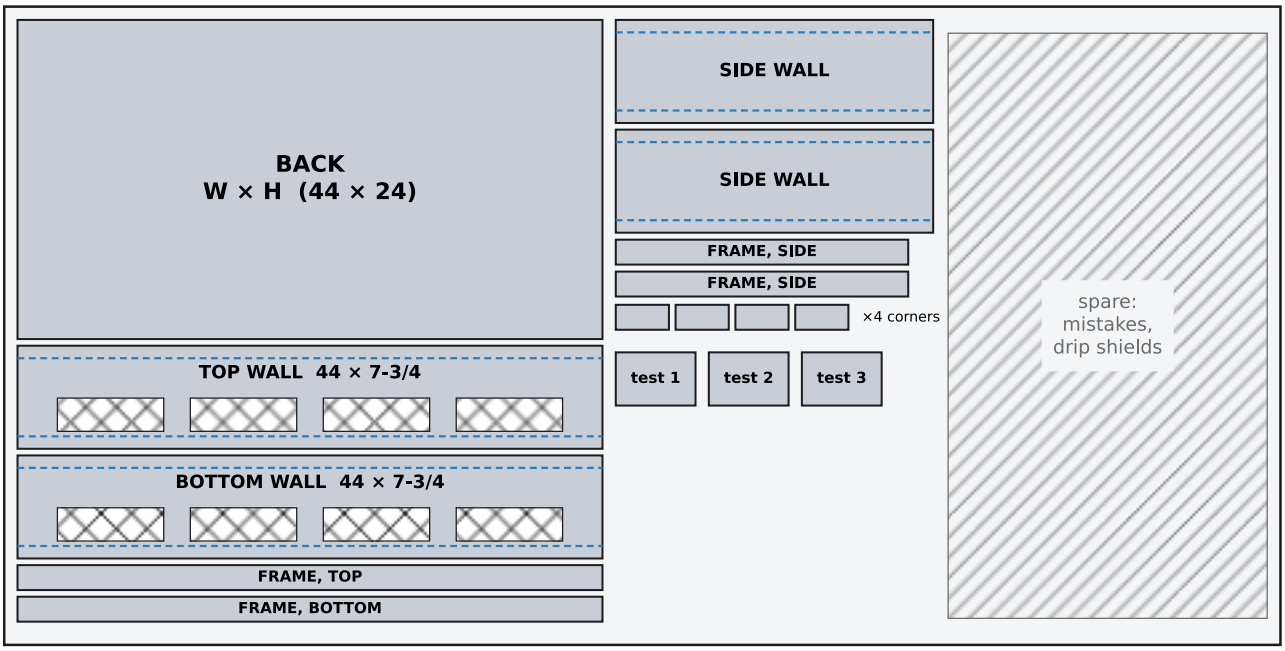
> **If every coupon cracks**
>
> The sheet may be 6061-T6 and not 5052 (5051 is an unusual label, so check). Stop scoring. Use a brake with a 1/8 in. inside radius, with a shallow 0.005 in. score only as a marker. The rest of this guide still applies.

2 Draw the flat parts and lay them out

2 h

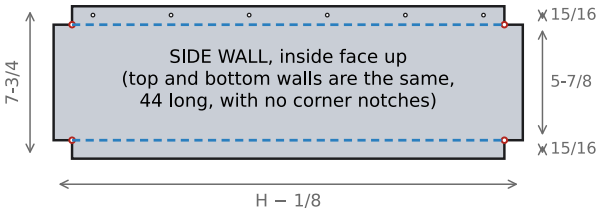
your measurements | CAD/CAM software

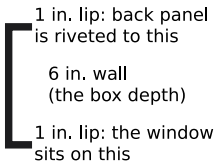
One 48 × 96 in. sheet, 1/16 in. thick (example sizes for a 44 × 24 in. window)



score line (V-bit, inside face) vent window, through-cut

End view once folded





Red circles: 1/8 in. relief holes where notches meet score lines, so the fold cannot tear into the corner.

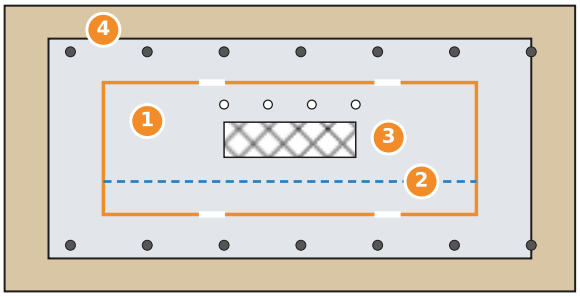
1. Draw each flat part from the table, on separate layers: **holes, scores, windows, outlines**.
2. **Rivet pilots:** 1/8 in. plunges, centred 15/32 in. from the edge on every back lip and around the matching edge of the back, every 4 in.; 3 per leg on the corner pieces.
3. **Windows:** four 8 × 2-1/2 in. in each of the top and bottom walls, positions in step 10's drawing, plus the small vent-screw pilots around them.
4. **Back panel:** four 5/16 in. circles for the VESA bolts (600 × 400 mm, centred), and pilots for the PSU, Pi and earth bolt.
5. **Bottom wall:** four 1/8 in. weep holes and a 13/16 in. gland circle in the gap between windows 3 and 4.
6. Tell the CAM to cut on the **inside face**: the scores must land inside the fold.

3 Cut on the CNC

2 to 3 h

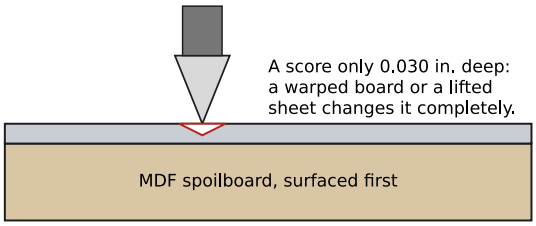
sheet | spoilboard, screws | O-flute bit | V-bit | WD-40

Top view: sheet on a surfaced MDF spoilboard



1 holes 2 scores 3 windows 4 outlines, with small tabs
Screws go through waste, never through a part.

Side view: why the board must be flat



Surface the board, then use a touch-off plate on top of the sheet.

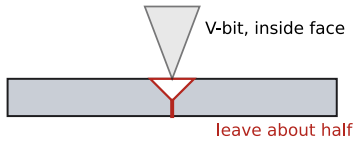
1. **Empty the dust collector first.** Aluminum chips in wood dust are a fire risk. Wear eye protection.
2. Surface the spoilboard, then screw the sheet down through waste areas, every 6 to 8 in. Thin sheet lifts, and lifted sheet ruins scores.
3. Cut in the order shown: **holes, scores, windows, outlines**. Outlines come last so the sheet stays held down until the end.
4. Starting point for the 1/8 in. O-flute: 18,000 RPM, 30 in./min, 0.020 in. per pass, three or four passes, with WD-40 on the cut. Listen and adjust: gummy chips mean too slow a feed.
5. Leave small tabs on outlines. Knock parts free, then file tabs flush and deburr every edge. Sharp aluminum cuts hands and wires.

> **Vacuum tables**
> Do not spray lubricant on a vacuum table: it pulls through to the pump.

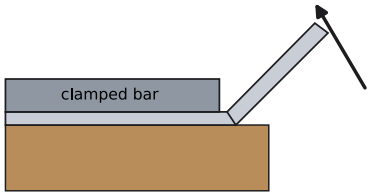
4 Fold the walls, frame and corners

90 min

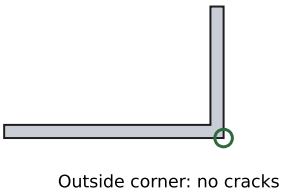
cut parts | steel angle or hardwood bar | C-clamps | square



1 Score on the INSIDE of the fold, in two light passes, tool centred on the line.



2 Score faces up, on the edge of the bar. Clamp every 8 in. One slow push to 90°.



3 Never unbend and refold. A fold that opens is scrap.

1. Lay a wall on the bench, score up, with the score on the edge of the bench bar. Clamp a second bar over it every 8 in.
2. Push the lip up to 90° in one slow, even motion, using a straight board across the whole length so it does not kink.
3. Fold the two lips on each wall, all the same way, so the wall is a C. Check each wall is straight.
4. Fold the clamp frame pieces and the four corner pieces. Bend the corner pieces in a vise without scoring: they carry load.

> **How strong is a scored fold?**
> Only as strong as the metal left under the groove. That is fine here: the lips are held flat by the riveted back and foam, and the back panel carries the load that would rack the box.

5 Cut the thick bars

30 min

2 sticks of 1/8 in. angle | hacksaw or mitre saw | file

A 1-1/2 × 1-1/2 × 1/8 in. angle, 8 ft



B 1-1/2 × 1-1/2 × 1/8 in. angle, 8 ft



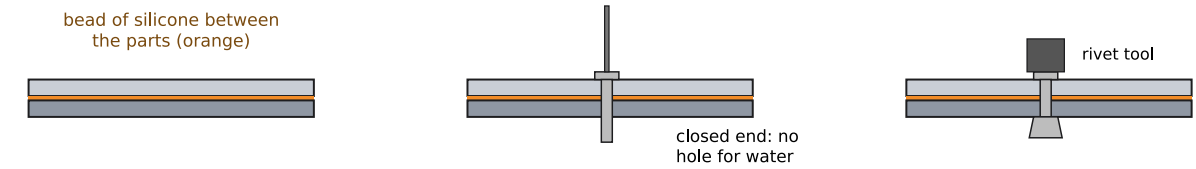
These four bars carry the weight. They are the only metal you cut with a saw. Lengths in inches.

1. Cut the two rails (W – 1/2) and the two backing bars (28 in.). Every end is square.
2. File each cut end until it no longer catches a fingernail, and label each bar.

6 Learn the sealed rivet

15 min

2 scraps | 3 rivets | silicone | rivet tool | drill, #20 bit



1 Bead of silicone on one part, then clamp the parts so the holes line up.

2 Push in a closed-end rivet from the outside. Flange flat.

3 Squeeze until the pin snaps. Silicone should show around the seam.

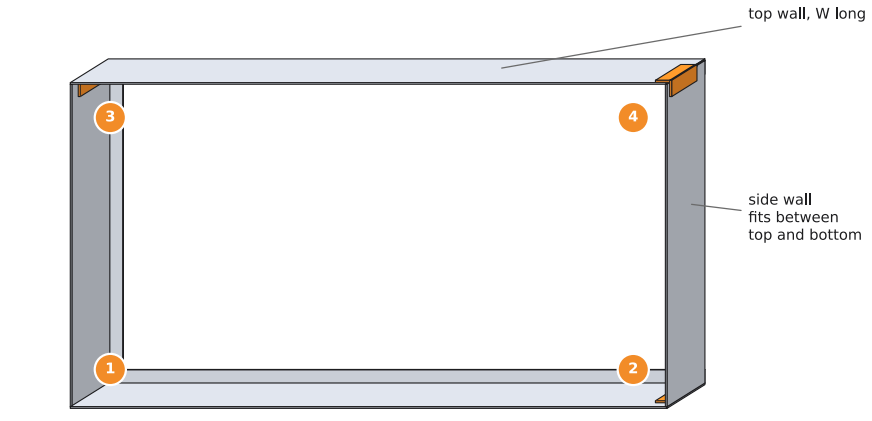
1. Clamp the two parts together. Open the 1/8 in. pilot to **#20 (0.161 in.)** through both at once, so the holes line up perfectly.
2. Unclamp, run a thin bead of silicone between the parts, and clamp again.
3. Push a rivet in from the outside, flange flat, and squeeze until the pin snaps. Do three on scrap first.

> **Spacing**
> Every 4 in. on long joints. If a hole misses, drill a fresh one beside it and rivet both. Wipe the squeeze-out before it skins.

7 Build the rim

60 min

4 walls | 4 corner pieces | 24 rivets | silicone | rivet tool | square | clamps



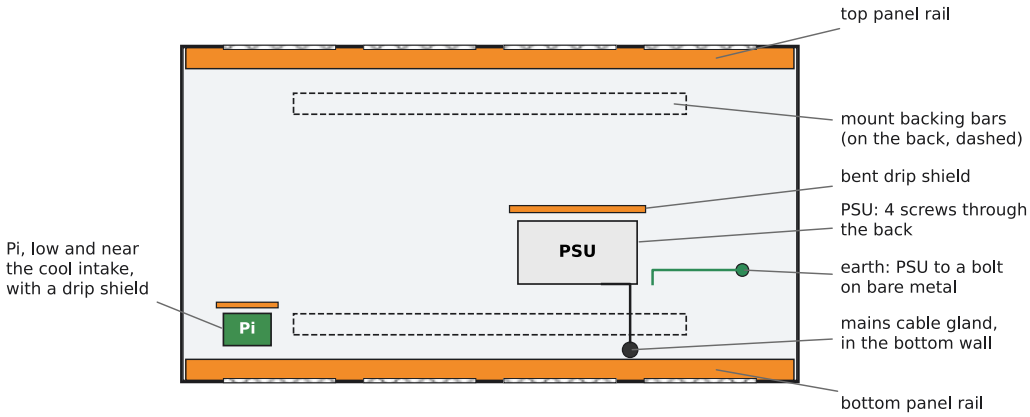
Front lips left out of this picture so you can see the four corner pieces.

1. Stand the four walls on edge, lips facing inward. The sides fit **between** the top and bottom walls.
2. Put a corner piece inside each corner (1 to 4) with silicone on both faces, and clamp.
3. Check each corner is square, then drill and rivet: 3 rivets into each wall at each corner.
4. Measure both diagonals. If they differ by more than 1/8 in., rack the box square before the last rivet goes in.

8 Rivet in the panel rails

40 min

2 rails | 12 rivets | the panel array | rivet tool | clamps

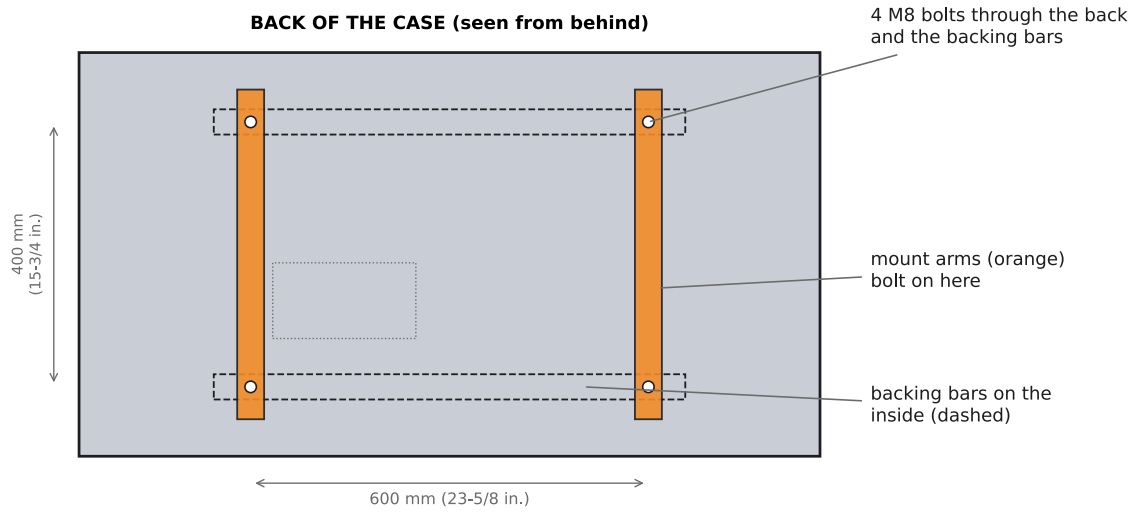


1. **Dry-fit first.** Set the array inside and find where its backplate brackets can bolt to a rail across the top and one across the bottom. The LED faces should end up **about 1/4 in. behind the front lips**.
2. Rivet each rail to the inside of the top or bottom wall: flat leg against the wall, running back from 1-1/4 in. behind the front edge, the other leg standing to take the backplate bolts. Six rivets per rail.
3. Keep the rails clear of the vent windows and of the well-nut zone in the front 1 in. of each wall.
4. Drill the rails for the backplate bolts now, while you can still reach.

9 Prepare the back panel

60 min

back | 2 backing bars | 4 M8 bolts, lock nuts, washers | PSU | drill



1. Clamp the two backing bars inside the back, one on the top pair of VESA holes and one on the bottom pair. Drill through the bars from the panel's holes.
2. Bolt on each mount arm from outside with an M8 bolt, washers, and a lock nut inside. Silicone under each washer. Check the arms hang the right way up.
3. Fix the PSU with M4 screws through the back, with a dab of silicone under each head.
4. Fix the Pi's nylon standoffs, low and left, near the cool intake.
5. Fit the earth bolt: #10 stainless screw through bare metal with a star washer. Scrape any coating off first.

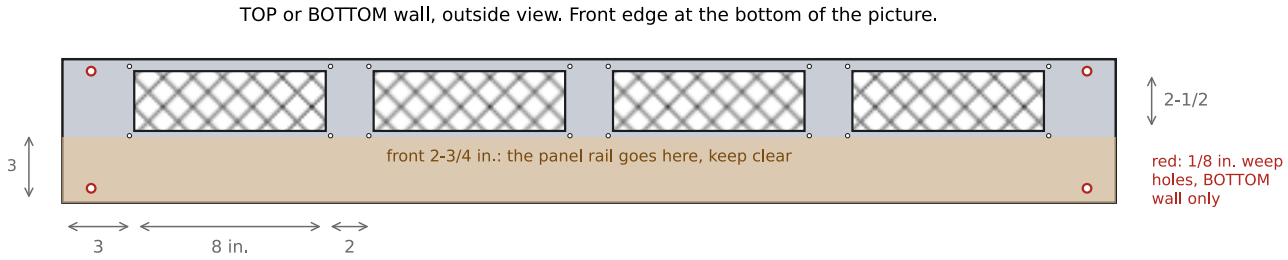
Which way is up

Hold the back as it will hang before you drill anything. Mount arms have a top and a bottom.

10 Print and fit the vents, fit the gland

30 min, plus printing

8 printed vents, 8 frames | insect screen | silicone | cable gland | screwdriver | scissors



1. Print eight vents in ASA: at least 4 perimeters, 30 % infill, with an enclosure so the part does not warp. Allow for the 1/16 in. radius the CNC leaves in window corners.
2. Cut eight patches of screen 1/2 in. larger than each window, and trap each between its vent and its inner frame.
3. Run a thin bead of silicone under each vent's flange and screw it in from outside with stainless screws. On the top wall, tilt the slats so openings face front. On the bottom wall, let them face down.
4. Fit the cable gland in the bottom wall, nut inside.

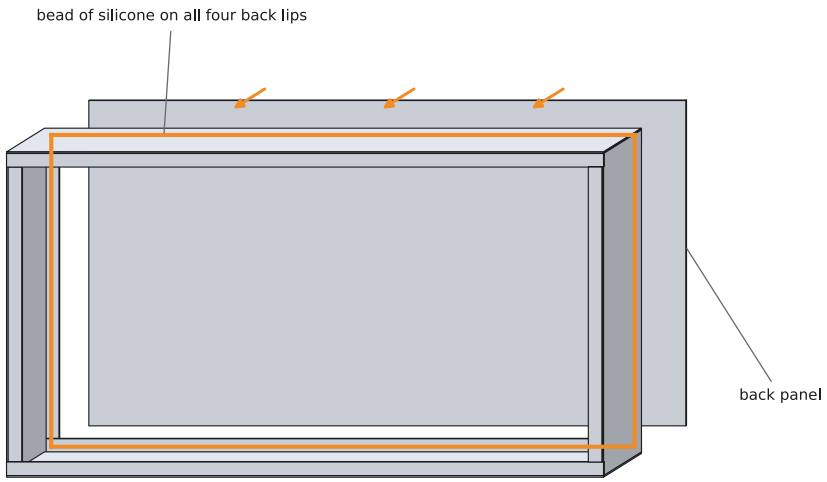
Why this much vent

The board normally gives off about 50 W of heat. Each wall has four 20 sq in. windows. Louvres and mesh take roughly two thirds of that, which still leaves enough opening for warm air to rise out on its own.

11 Fit the back

60 min

rim | back panel | about 34 rivets | silicone | rivet tool



1. Clamp the back onto the rim and open the pilots to #20 through both. Unclamp and clean out chips.
2. Run a continuous bead of silicone along all four back lips.
3. Press the back on and rivet every 4 in., working out from the middle of each side. Wipe the squeeze-out.

12 Electronics, coating and drip shields

90 min

Pi + bonnet | standoffs | earth wire | 16 AWG red/black | conformal coating | 2 drip shields | screwdrivers | wire strippers

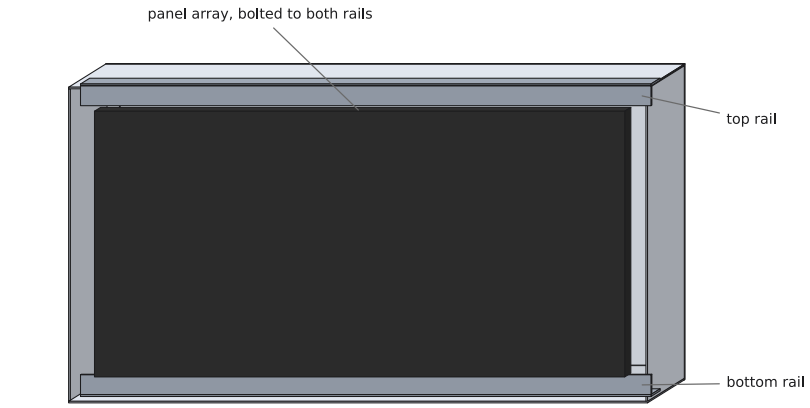
1. **Coat the Pi and bonnet** off the board. Mask every connector, the SD slot, USB and Ethernet ports, GPIO pins and the fan. Two light coats, and let them cure fully (see the can) before powering on.
2. Fit the Pi with its Active Cooler, riser and 16 mm standoffs, then the bonnet.
3. Fold and screw the two scrap drip shields above the PSU and Pi, tilted so water runs off to the side, with a gap so air still reaches them.
4. With the cord **unplugged**, run it in through the gland with a drip loop outside, and tighten the gland.
5. Connect green earth wire from the PSU's earth terminal to the earth bolt.
6. Give the Pi its own 16 AWG pair straight from the PSU terminals (sheet 4 of the wiring document).

> **Mains**
> Only the cord passes through the gland; the PSU's mains connections stay as they are. Anything on the 120 V side is an electrician's job.

13 Panels in, first light

45 min

panel array · backplate bolts · wrench



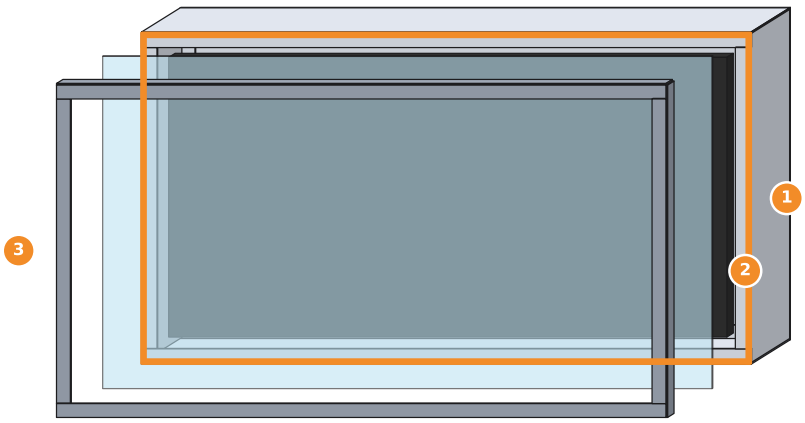
Front lips left out so the rails show.

1. Lower the panel array in through the front and bolt its backplates to both rails. Route ribbons and power away from sharp edges.
2. Plug in and run the board. Check every panel lights, the right way up, before you close anything.
3. Leave it on for 20 minutes and look for flicker on a white screen (the power test in `HARDWARE.md`).

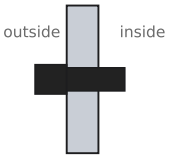
14 Window and clamp frame

60 min

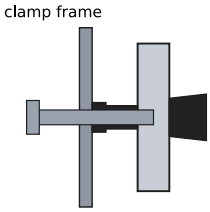
plastic sheet | foam tape | clamp frame (4 pieces) | 12 well nuts | 12 security screws | drill, 3/16 + 3/8 in. | T-25 pin bit



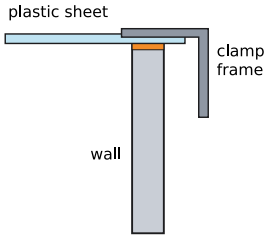
1 foam tape on the front lips 2 your plastic sheet 3 clamp frame, screwed into well nuts



1 Push the well nut into a 3/8 in. hole from outside. Flange stays outside.



2 The screw passes through the clamp frame; the rubber bulges behind and grips. Watertight.



3 Cut-away: the sheet is squeezed between two strips of foam tape (orange).

1. Stick foam tape all round the front lips (1). Clean the plastic sheet and peel any film.
2. Lay the sheet on the tape (2). Stick foam tape inside each clamp-frame piece's front leg.
3. Set the clamp frame on (3): top and bottom pieces full width, side pieces between them. Clamp it.
4. Drill 3/16 in. through the frame's side leg and the wall, about 5/8 in. back from the front edge: 4 holes along the top, 4 along the bottom, 2 down each side.
5. Take the frame off. Open just the **wall** holes to 3/8 in. and push a well nut into each from outside. Refit the frame and drive the screws until the foam squeezes about halfway. **No tighter:** acrylic cracks if clamped hard.

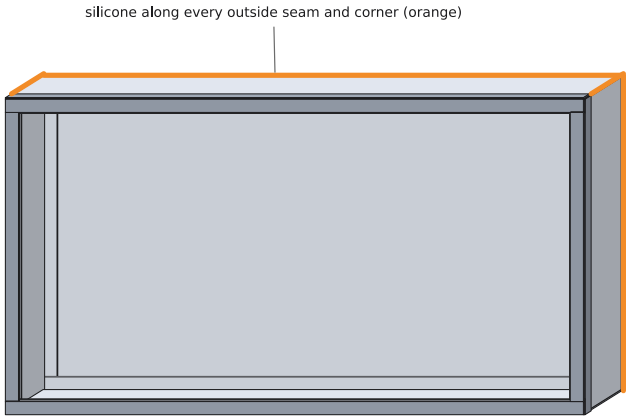
No holes in the plastic

Nothing goes through the sheet. It floats between two strips of foam, so it can grow and shrink with the seasons without cracking.

15 Seal the outside

30 min, then overnight

silicone | caulk gun | masking tape, rag



1. Mask both sides of each outside seam. Run a thin bead where the back meets the walls, and down each corner.
2. Smooth it with a wet finger, pull the tape, and let it cure overnight.
3. **Do not** silicone the clamp frame or any vent or weep hole.

16 Test before it goes up

3 h, mostly waiting

Test	How	Pass
Earth	Multimeter from the plug's earth pin to a wall of the case and to the PSU case	Near 0 Ω (it beeps)
Rain	Garden hose on the shower setting, on the top, sides and window for 10 minutes. Then open the frame and look.	Dry inside
Heat	Closed up, normal board, 2 hours. Then `vcgencmd measure_temp`	Pi under 75 °C
Power	`vcgencmd get_throttled` after a white-screen test	`0x0`
Weight	Bathroom scale	Write it on the back

Hanging it

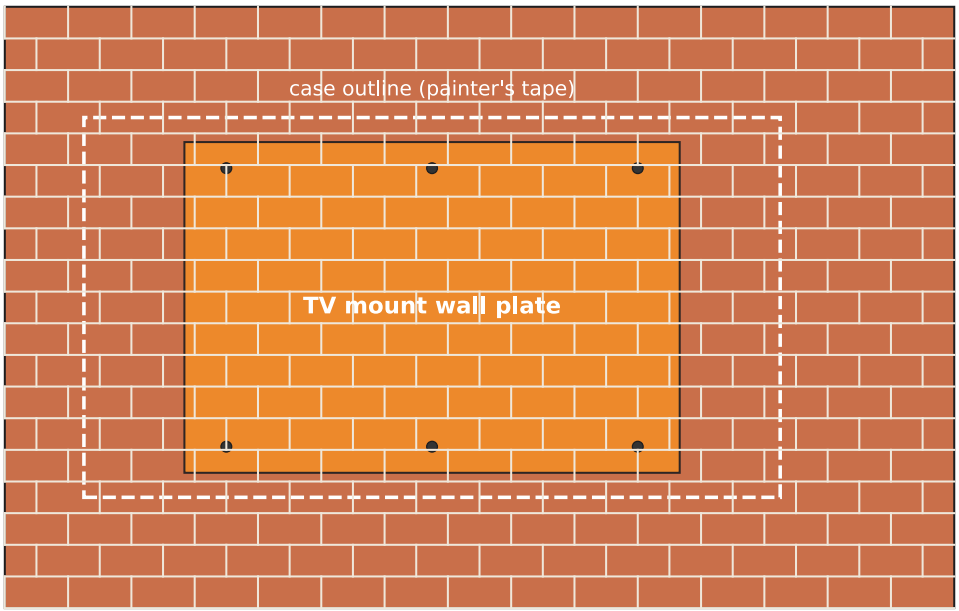
Before you drill

SEPTA's OK in writing; two people; two ladders or a platform. Drill one test hole low and out of sight first. If the dust comes out red and the hole stays crisp, the brick is solid and sleeve anchors are fine. If it is crumbly or breaks into a hollow, stop: use the epoxy method in `docs/mounting-and-case-plan.pdf` instead.

A Mark the wall

20 min

wall plate · level · painter's tape · tape measure



6 anchors, each in the middle of a whole brick. Use the plate's slots that land best.

Level the plate before the last anchor is tightened.

1. Tape the case outline on the wall, bottom edge 7 to 8 ft above the platform.
2. Hold the plate centred inside the outline and level. Mark 6 slots, each landing in the middle of a whole brick (or in mortar joints, if SEPTA prefers).

B Anchors and the wall plate

45 min

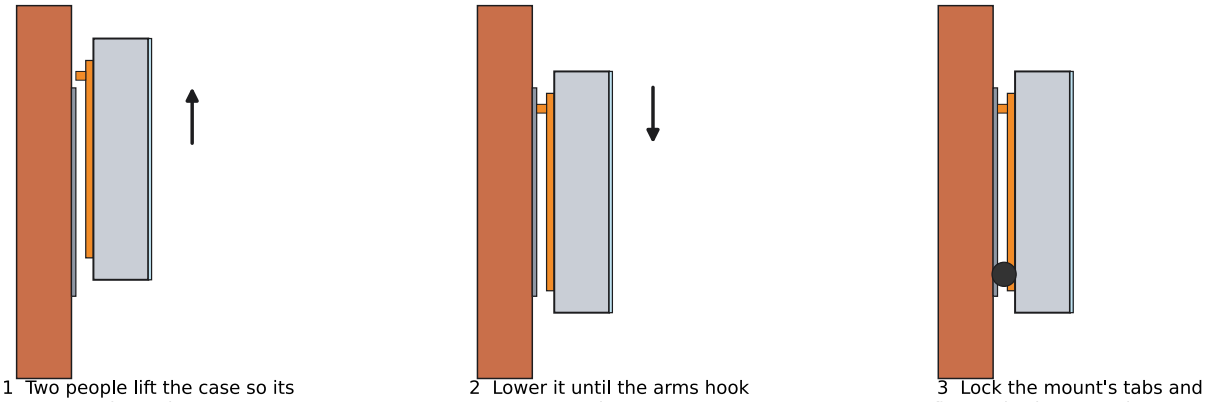
6 sleeve anchors · fender washers · drill, 1/4 in. masonry bit · hammer · wrench

1. Start each hole with the drill on **rotary** (hammer off) so the brick face does not chip, and go 1/2 in. deeper than the anchor.
2. Blow the dust out. Tap each anchor in through the plate.
3. Fender washer under each nut. Tighten until snug, then one more turn, and check the plate is still level.

C Hang it and lock it

30 min

the case · padlock or security screw · 2 people



1. Two people lift the case until its arms are just above the plate, then lower it until they hook on.
2. Level it with the mount's adjustment screws if it has them.
3. Rotate the mount's locking tabs and fit a padlock or security screw, so it cannot be lifted off.
4. Plug into the GFCI outlet (fitted by the electrician) with a drip loop, and press TEST, then RESET.

The cheaper route: a Z-bar

Save about $160 by using a 30 in. Hangman HM-30D Z-bar instead of a TV mount. Bolt the case half high on the back through one 32 in. 1/8 in. backing bar, hook pointing **down**. Fix the wall half to the brick with 4 anchors, hook pointing **up**. Then add two 2 in. long pieces of 1-1/2 in. angle at the bottom, one leg anchored to the wall and one under the case with a security screw, so it cannot be lifted off.

Looking after it

When	Do this
You need to get inside	Undo the 12 clamp-frame screws (T-25 pin bit) and lift off the frame and sheet. Everything is reachable from the front once the panel array is unbolted.
Water inside after rain	Look for a missed silicone seam at the back corners, then for flattened foam at the window. Re-seal; replace foam that has taken a set.
Droplets inside	Condensation, not a leak. Check the vents are clear and the board has stayed powered. Check the weep holes are open.
It runs hot in summer	Clean the mesh. If it is still hot, the full plan adds a small fan at the top vent.
Window scratched or cloudy	Mild soap and water only. No glass cleaner: ammonia ruins acrylic and polycarbonate.
Taking it down	Two people: unlock the mount's tabs, lift up, pull away.

Every spring

- ☐ Clean the window and the vent mesh; check the weep holes
- ☐ Look over the silicone and the foam; re-seal anything cracked
- ☐ Check the anchors and the mount locks are tight
- ☐ Press TEST and RESET on the GFCI outlet

Source files

Built from `docs/simple-build/` in github.com/BenGreenberg07/septa-tracker: `guide.html` for the words, `figures.py` for the pictures, and `./build.sh` to make the PDF. The heavier-duty design is `docs/mounting-and-case-plan.pdf`.